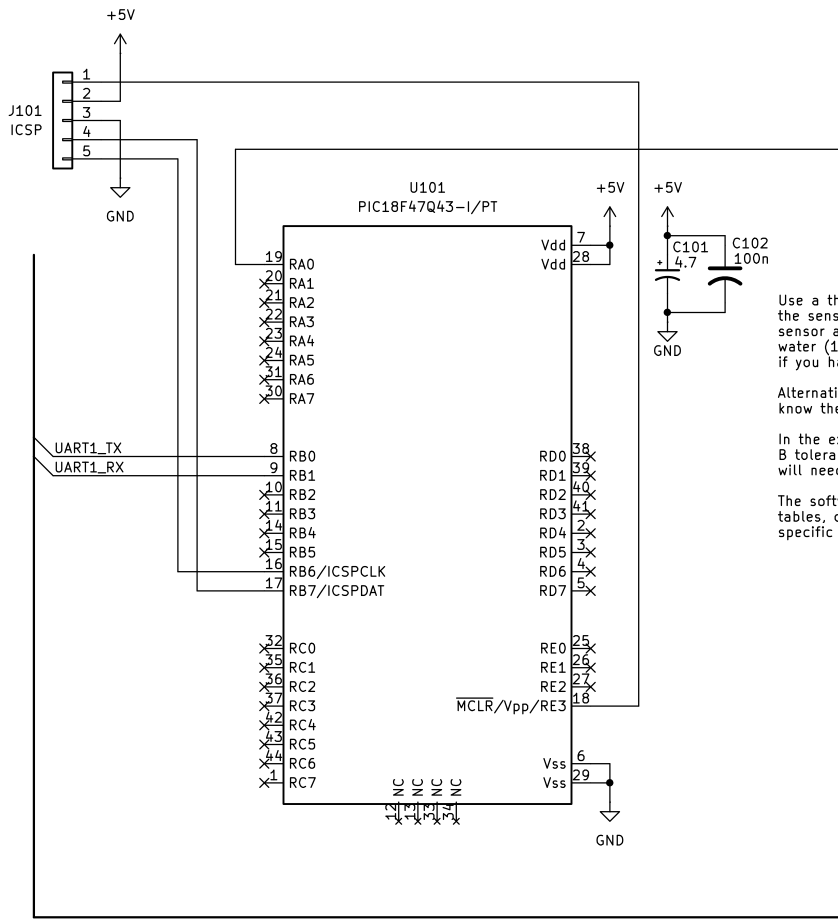
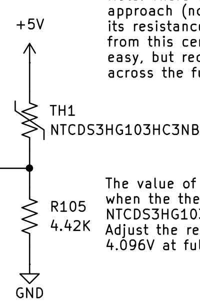


Note: There are several more ways to linearize a thermistor that are not shown. One approach (not shown) is to use a resistor in parallel with the thermistor that corrects its resistance at the center of the range your interested in. As you get further away from this central correction, the error will increase. This approach is convenient and easy, but reduces the dynamic range of the thermistor and does not provide true linearization across the full scale.



The value of R105 is chosen to produce a voltage at the junction of less than 4.096V when the thermistor is at it's lowest resistance. For the values shown, using TDK NTCDS3HG103HC3NB, the range of voltages from 0C to 100C are approx 0.69V to 4.08V. Adjust the resistor for the thermistor you are using to produce a voltage less than 4.096V at full scale.

Use a thermistor that reads about 10K at room temperature and has a negative temperature coefficient. This allows the sensed voltage to increase as the temperature increases (the NTC thermistor decreases in value). Calibrate the sensor at several temperatures if possible, such as freezing (0C or 32F), room temperature (25C or 77F), and boiling water (100C or 212F). This 3 point calibration should provide a good linearization of the thermistor. Alternatively, if you have access to the linearization tables of the thermistor you are using, these can be entered directly.

Alternatively, you can also use the thermistors "beta" value and the software can correct the temperature reading, if you know the beta of the thermistor you are using.

In the example circuit I used a TDK NTCDS3HG103HC3NB. The beta for this part is 3400K, has a resistance of 10K at 25C, and a B tolerance of 2%. It is a relatively inexpensive part. You can use any thermistor you like (including PTC) but the software will need to be adjusted appropriately.

The software has different symbols that can be defined to demonstrate using the calibration points, the linearization tables, or the beta values and tolerances. The code must be compiled with one of these macros enabled to demonstrate a specific approach.